

PROFESSIONAL SUMMARY

AI infrastructure engineer with nearly five years of cloud-native AI/HPC experience, focused on production LLM serving since 2024. Leads a Kubernetes-native MaaS control plane for heterogeneous non-NVIDIA accelerators, delivering Dynamo Runtime, a P/D Planner, the KV Event Plane, Mooncake-based RDMA KV transfer, KV-aware routing, and production model fast loading; also builds BatchFlow, LiteLLM/Higress gateways, autoscaling, failure recovery, and observability. Strong in Go and Python, with Rust engineering experience.

EXPERIENCE

Zhejiang Lab

Aug 2021 - Present

Senior Research Specialist · High-Performance Computing Facility Research Center

2024 - Present

LLM Serving / Model-as-a-Service (MaaS) Platform

- Led a production LLM-serving platform and Kubernetes Operator control plane, integrating SGLang/vLLM workers on heterogeneous non-NVIDIA accelerators with Dynamo DistributedRuntime and standardizing model declarations, worker lifecycle, P/D topology, layered readiness, and failure isolation.
- Connected worker bootstrap, the KV Event Plane, the Mooncake KV data plane, and a KV-aware Router; productionized ModelExpress-based tiered model loading so request routing, KV reuse, and elastic scale-out operate as one runtime path.
- Built a production P/D Planner that derives prefill/decode replica targets from request shapes, KV hits, and Capacity Profiles, then executes auditable ScaleIntents with readiness gates, stabilization windows, cooldowns, and rollback protection.
- Delivered BatchFlow multi-tenant offline inference, LiteLLM/Higress gateways, and a Rust Router; owned releases, rollbacks, and production diagnosis across queueing, routing, transfer, and model loading using request IDs, Prometheus metrics, and Kubernetes state.

Engineering Specialist · Intelligent Supercomputing Research Center

Aug 2021 - 2023

Cloud-Native AI / HPC Platform

- Customized Kubernetes Device Plugin and resource-management paths for heterogeneous accelerator discovery, node resource reporting, container-runtime validation, device-level allocation, and fault isolation.
- Adapted PyTorch workloads, models, and distributed job environments to Linux distributions developed in China and heterogeneous accelerator stacks, establishing the platform foundation for later serving and scheduling work.

SELECTED PRODUCTION SYSTEMS

Dynamo Runtime and ModelExpress-based Tiered Model Loading

DistributedRuntime · Tiered Cache · Readiness

- Productionized SGLang/vLLM runtimes on Dynamo DistributedRuntime, unifying worker lifecycle, Namespace/Component/Endpoint discovery, ModelRuntimeConfig/ModelDeploymentCard registration, P/D bootstrap, and graceful drain; enforced layered readiness across process, model, KV events, transport, and Router registration.
- Productionized ModelExpress-based tiered loading with immutable compatibility keys spanning model revision, dtype, quantization, TP/EP topology, and vendor runtime; selects compatible READY worker device memory, node-local NVMe, or CPFS/NFS and combines prefetch, pre-sharded checkpoints, parallel loading, and cache-aware placement to reduce scale-out cold-start overhead.
- Enforced lease-based de-duplication, checksum-gated atomic publication, and vendor JIT/operator/graph cache reuse; admitted traffic only after weight validation, warmup, health checks, KV events, and Router registration, with fallback to NVMe, shared storage, or the native loader.

KV Event Plane and Mooncake KV Data Plane

KV-aware Routing · RDMA KV Transfer · Fallback

- Consumed per-DP-rank SGLang/vLLM BlockStored, BlockRemoved, and AllBlocksCleared events into the Event Plane and maintained a Worker/Rank/Storage-Tier prefix radix index.
- The Router selects P/D workers from prefix KV overlap, load, accelerator/KV-layout compatibility, and network topology; Mooncake Transfer Engine performs batched KV tensor transfers through the high-performance RDMA fabric, degrading to load-based routing on event anomalies and isolating failed pairs before reselection and retry.

SELECTED PRODUCTION SYSTEMS (CONTINUED)

Adaptive P/D Planner and Autoscaling Guardrails

[Capacity Profile](#) · [ScaleIntent](#) · [Guardrails](#)

- Aggregates request rate, ISL/OSL, prefill compute tokens and cached tokens, KV hits, queue depth, and running state over rolling windows; selects Capacity Profiles by model, accelerator, quantization, TP/DP, and engine version, using interpolation and conservative fallback for safe capacity estimates.
- Computes prefill/decode token demand, safe per-worker throughput, and independent replica targets, then derives an aggregate target from the bottleneck stage; continuously emits auditable ScaleIntents with request shape, capacity samples, utilization, and blocking reasons.
- Decouples the Planner, ScaleIntent Controller, ScalingAdapter, and IGD/ISI/Scheduler; applies desired > ready pending gates, one-step changes, stabilization windows, cooldowns, stale-intent rejection, and rollback to suppress oscillation and concurrent writes.

KV-aware Router, Multi-Tenant Admission, and Tracing

[P/D Routing](#) · [Priority Admission](#) · [Tracing](#)

- Integrated SGLang HTTP/gRPC P/D routing and selects compatible prefill/decode workers from KV overlap, worker load, and network topology; propagates request IDs, bootstrap/transfer context, and KV telemetry for cross-component diagnosis.
- Implemented P0-P3 priority admission in Rust with global/per-priority semaphores, bounded queues, strict priority, tenant round-robin, and wait timeouts; holds permits until streaming response bodies complete so capacity is not released early.

BatchFlow Multi-Tenant Offline Batch Inference

[OpenAI Batch API](#) · [Scheduling](#) · [Recovery](#)

- Models each batch as a durable, recoverable asynchronous job: the database is the system of record, while schedulers/workers handle transactional claims, sharding, lease renewal, and finalization; Kubernetes Leases, custom_id de-duplication, and shard recovery support multiple replicas.
- Delivered OpenAI Batch API-compatible streaming JSONL, tenant/model concurrency budgets, cancellation, expiry cleanup, output/error writeback, and request-level error isolation to bound offline pressure on shared online model backends.

CORE TECHNOLOGIES

Runtime & Data Plane: Dynamo Runtime / DistributedRuntime, SGLang, vLLM, KV Event Plane, Mooncake Transfer Engine, RDMA KV transfer, P/D disaggregation

Control Plane & Scheduling: Go, Kubernetes Operator, CRD/controller, Capability/Capacity Profiles, ScaleIntent, P/D autoscaling, Device Plugin, Helm, ArgoCD

Model Loading & Routing: ModelExpress, CPFS/NFS, node-local NVMe, pre-sharded checkpoints, cache-aware scheduling, KV-aware routing, Rust

Serving & Observability: Python, OpenAI-compatible API, Batch API, LiteLLM, Higress, Redis, MySQL, Prometheus, QPS/TTFT/TPOT, request tracing

EDUCATION

Arizona State University - M.S. in Computer Science, GPA 3.97 / 4.00, May 2019 - Dec 2020

Arizona State University - B.S. in Computer Science, Aug 2015 - May 2019

PUBLICATION & INTELLECTUAL PROPERTY

Prompt2Length: A Lightweight Model for Predicting Output Length in LLM Inference - third author, IEEE AIAHPC 2025; co-inventor on two granted invention patents and named contributor on two registered software copyrights.

LANGUAGES

Mandarin Chinese (native); English (full professional proficiency).